AA–TV wiring v6 (architecture v5): two solder–boards, waterproof connectors, max 2 penetrations per drybox

BUOY (surface, dry)

5V_PI
BT2
USB–C power bank 5 V / 3 A
GND_PI

J1

GPIO12/PWM0 — 5V — 2 — 5V_PI
GPIO13/PWM1 — 3V3 — 1
GPIO14/UART_TXD — ID_SD_I2C0/GPIO00 — 27
GPIO15/UART_RXD — ID_SC_I2C0/GPIO01 — 28
GPIO16/SPI1_CE2 — SDA_I2C1/GPIO02 — 3
GPIO17/SPI1_CE1 — SCL_I2C1/GPIO03 — 5
GPIO18/SPI1_CE0/PCM_CLK/PWM0 — GPCLK0/GPIO04 — 7
GPIO19/SPI1_MISO/PCM_FS — GPCLK1/GPIO05 — 29
GPIO20/SPI1_MOSI/PCM_DIN/PWM1 — GPCLK2/GPIO06 — 31
GPIO21/SPI1_SCLK/PCM_DOUT
GPIO22/SDIO_CLK — CE1_SPI0/GPIO07 — 26
GPIO23/SDIO_CMD — CE0_SPI0/GPIO08 — 24
GPIO24/SDIO_DAT0 — MISO_SPI0/GPIO09 — 21
GPIO25/SDIO_DAT1 — MOSI_SPI0/GPIO10 — 19
GPIO26/SDIO_DAT2 — SCLK_SPI0/GPIO11 — 23
GPIO27/SDIO_DAT3
GND — 6

Raspberry Pi 4 (buoy)

Pi 4 powered from its own power bank (USB–C).
Pi USB–A port feeds W1: 5 V + USB data to the Pico.
Pi side: /dev/ttyACM0 (mpremote, pyserial).
Hard reset of the Pico = power–cycle the Pi USB ports (uhubctl), since the RUN wire is gone.

TETHER

W1
USB–A — micro–B
5V_PI — 1 — 5 — VBUS_PICO
GND_PI — 4 — 8 — GND
+ USB data

PEN1
A1: air–tube bulkhead (hose barb)

USB–A to micro–B, ~3 m

Vinyl air tube from the buoy.
The USB cable runs inside it.

MAIN DRYBOX (upper, on hull)

BOARD A: Pico solder–board

GP12 — R1 220 — M4_PWM
GP13 — R2 220 — M5_PWM
GP14 — R3 220 — M6_PWM
GP15 — R4 220 — M7_PWM
GP16 — R5 220 — THR_PWM
GP17 — R6 220 — DRV_L_PWM
GP18 — R7 220 — DRV_R_PWM
GP19 — R8 220 — SPARE_PWM

220 ohm in series on every PWM line that leaves the drybox (GPIO protection if a wet connector shorts).

J2
Leak sensor (main box)
1 +3V3
2 GND
3 LEAK_MAIN

VBUS_PICO
+3V3
Raspberry Pi Pico
A1

VSYS — 39
VBUS — 40
3V3 — 36
RUN — 30
3V3_EN — 37
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12 — 16
GPIO13 — 17
GPIO14 — 19
GPIO15 — 20
GPIO16 — 21 — GP16
GPIO17 — 22 — GP17
GPIO18 — 24 — GP18
GPIO19 — 25 — GP19
GPIO20 — 26
GPIO21 — 27
GPIO22 — 29
ADC_VREF — 35
GPIO26_ADC0 — 31 — LEAK_MAIN
GPIO27_ADC1 — 32 — LEAK_REAR
GPIO28_ADC2 — 34
AGND — 33 — GND
GND — 3
GND

Pico: VBUS from the Pi over USB.
VSYS unused (no BEC fed any more).
GP0/GP1 UART and RUN unused.

J3 — P1

1	M4_PWM	1	M4_PWM
2	M5_PWM	2	M5_PWM
3	M6_PWM	3	M6_PWM
4	M7_PWM	4	M7_PWM
5	THR_PWM	5	THR_PWM
6	DRV_L_PWM	6	DRV_L_PWM
7	DRV_R_PWM	7	DRV_R_PWM
8	SPARE_PWM	8	SPARE_PWM
9	LEAK_REAR	9	LEAK_REAR
10	+3V3	10	+3V3
11	GND	11	GND
12	GND	12	GND

Screw terminal 12–pos (to H1)
A2: IP68 12–pin panel receptacle

HARNESS H1

H1: 12–core jacketed cable, IP68 12–pin plug each end (e.g. M12 A–coded 12–pin or Weipu SP21 12–pin). Pins: 1–4 servos M4–M7, 5 thruster, 6–7 wheel L/R, 8 spare (pump), 9 rear leak, 10 +3V3, 11–12 GND.

P2 — P3

1	M4_PWM	1	M4_PWM
2	M5_PWM	2	M5_PWM
3	M6_PWM	3	M6_PWM
4	M7_PWM	4	M7_PWM
5	THR_PWM	5	THR_PWM
6	DRV_L_PWM	6	DRV_L_PWM
7	DRV_R_PWM	7	DRV_R_PWM
8	SPARE_PWM	8	SPARE_PWM
9	LEAK_REAR	9	LEAK_REAR
10	+3V3	10	+3V3
11	GND	11	GND
12	GND	12	GND

H1 plug, end A
H1 plug, end B

REAR DRYBOX (inside open rear endcap)

BOARD B: servo / thruster logic + 7.5 V rail

J5 — Servo M4 header (JR) — 1 M4_PWM, 2 SERVO_7V5, 3 GND
J6 — Servo M5 header (JR) — 1 M5_PWM, 2 SERVO_7V5, 3 GND
J7 — Servo M6 header (JR) — 1 M6_PWM, 2 SERVO_7V5, 3 GND
J8 — Servo M7 header (JR) — 1 M7_PWM, 2 SERVO_7V5, 3 GND
J9 — Thruster ESC sig (yel/wht) — 1 THR_PWM, 2 GND
J10 — Wheel ESC signals (via R2) — 1 DRV_L_PWM, 2 DRV_R_PWM, 3 GND
J11 — Leak sensor (rear box) — 1 +3V3, 2 GND, 3 LEAK_REAR
J13 — Spare PWM (ballast pump) — 1 SPARE_PWM, 2 GND

J12
1 SERVO_7V5
2 GND
7.5 V in from buck

SERVO_7V5
C1
1000uF 16V
GND

Board B rules:
– No 12 V on the breadboard.
– 7.5 V + GND: 18 AWG bus wire along the strips.
– C1 next to the servo headers.

J4

1	M4_PWM	1	M4_PWM
2	M5_PWM	2	M5_PWM
3	M6_PWM	3	M6_PWM
4	M7_PWM	4	M7_PWM
5	THR_PWM	5	THR_PWM
6	DRV_L_PWM	6	DRV_L_PWM
7	DRV_R_PWM	7	DRV_R_PWM
8	SPARE_PWM	8	SPARE_PWM
9	LEAK_REAR	9	LEAK_REAR
10	+3V3	10	+3V3
11	GND	11	GND
12	GND	12	GND

R1: IP68 12–pin panel receptacle
Screw terminal 12–pos (from H1)

Off–board, in rear drybox

U1 — 12 V –> 7.5 V buck
VBAT — 1 — VIN+
GND — 2 — VIN–
VOUT+ — 3 — SERVO_7V5
VOUT– — 4 — GND

U2 — TD1.2 thruster ESC (12 V direct)
VBAT — 1 — BAT+
GND — 2 — BAT–
U — 5 — THR_U
V — 6 — THR_V
W — 7 — THR_W
THR_PWM — 3 — SIG
GND — 4 — SIG_GND
BLDC ESC

Thruster ESC has no BEC: only signal + signal ground go to Board B (J9). Its 12 V comes straight off the XT60 Y–pigtail, never the board.

J14/J15: plain XT60 pair (dry side).
J14 = battery cable end, J15 = Y–pigtail to thruster ESC U2 + buck U1.

J14 — VBAT, GND
J15 — VBAT, GND

PEN2
R2: multi–hole cable gland
R2 carries 7 cables in one hole: battery 2–core, thruster cable, 4 servo leads (round them with adhesive heat–shrink), wheel–signal 3–core.

WET ZONE (flooded hull, open rear endcap, splash areas)

J16/J17: Amass XT60W waterproof pair (battery side / harness side).

BT1
3S LiPo 5000 mAh
BAT_P — J16 1 — 1 J17 — BAT_P — F1 — VBAT
BAT_N 2 — 2
BAT_N — 30 A blade, waterproof holder
SW1
1 — 2
GND
Kill switch (low side)

S1: after F1/SW1, one soldered Y–splice (adhesive heat–shrink) splits VBAT/GND to:
– rear drybox cable (through R2) –> J14
– wheel ESC U3
– wheel ESC U4

U3
VBAT — 1 — BAT+
GND — 2 — BAT–
DRV_L_PWM — 3 — BEC+
GND — 4 — SIG
5 — SIG_GND
ESC
M+ — 6 — M1
M– — 7 — M1
Driven wheel L (550)
Wheel ESC L (waterproof)

U4
VBAT — 1 — BAT+
GND — 2 — BAT–
DRV_R_PWM — 3 — BEC+
GND — 4 — SIG
5 — SIG_GND
ESC
M+ — 6 — M2
M– — 7 — M2
Driven wheel R (550)
Wheel ESC R (waterproof)

Wheel ESCs live wet, so they must be a waterproof type.
SIG/SIG_GND: spliced to the 3–core cable from J10.
BEC+ (red signal wire): cut and seal, Pico has its own 5 V.

PENETRATIONS (target: 2 per drybox)
Main drybox: A1 air–tube bulkhead (USB cable inside, vents to buoy)
A2 IP68 12–pin panel receptacle P1 (harness H1)
Rear drybox: R1 IP68 12–pin panel receptacle P4 (harness H1)
R2 multi–hole gland PEN2 (battery, thruster, 4 servos, wheel signals)

POWER PATH
BT1 –> XT60W pair (wet) –> F1 (+) / SW1 (–) –> Y–splice S1 –> wheel ESCs U3/U4
–> 2–core through R2 –> XT60 J14/J15 (dry)
–> thruster ESC U2 (12 V) + buck U1 (7.5 V servos)

Board B carries only 7.5 V servo power and signals. Board A carries only logic.

GROUND: one GND. Pico GND reaches battery GND through H1 pins 11–12.
GND_PI and GND join through the USB cable shield/ground only.

Servo leads and the thruster cable go through gland R2 and plug in inside the rear drybox.

MT1
THR_U — 1
THR_V — 2
THR_W — 3
BLDC
TD1.2 thruster

M4_PWM — SERVO_7V5 — GND — PWM + – — M4 — Rudder servo
M5_PWM — SERVO_7V5 — GND — PWM + – — M5 — Rudder servo
M6_PWM — SERVO_7V5 — GND — PWM + – — M6 — Elevator servo
M7_PWM — SERVO_7V5 — GND — PWM + – — M7 — Elevator servo

Pico powered + data over USB from the buoy Pi 4. Servos on 7.5 V buck, thruster on 12 V. Follows architecture v5. No PCB: two solderable breadboards (Board A, Board B).

PIE AA–TV

Sheet: /
File: AATV.kicad_sch

Title: AA–TV electrical: solder–board split, connectors and drybox penetration

Size: A2	Date: 2026–11–03	**Rev: v6**
KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4		Id: 1/1